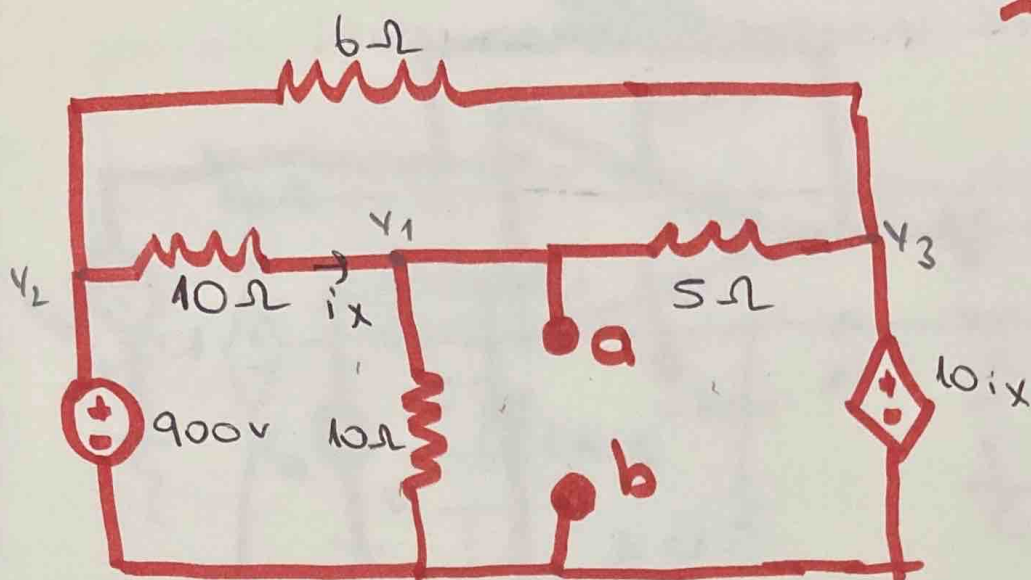


2020



$$V_2 = 900\text{V}, V_3 = 10i_x, V_1 = V_{Th}$$

$$i_x = \frac{V_2 - V_1}{10} = \frac{900 - V_1}{10} \rightarrow 10i_x = 900 - V_1$$

For node -1:

$$\frac{V_1 - 900}{10} + \frac{V_1}{10} + \frac{V_1 - 10i_x}{5} = 0$$

$$V_1 - 900 + V_1 + 2V_1 - 20i_x = 0$$

$$4V_1 - 20i_x = 900$$

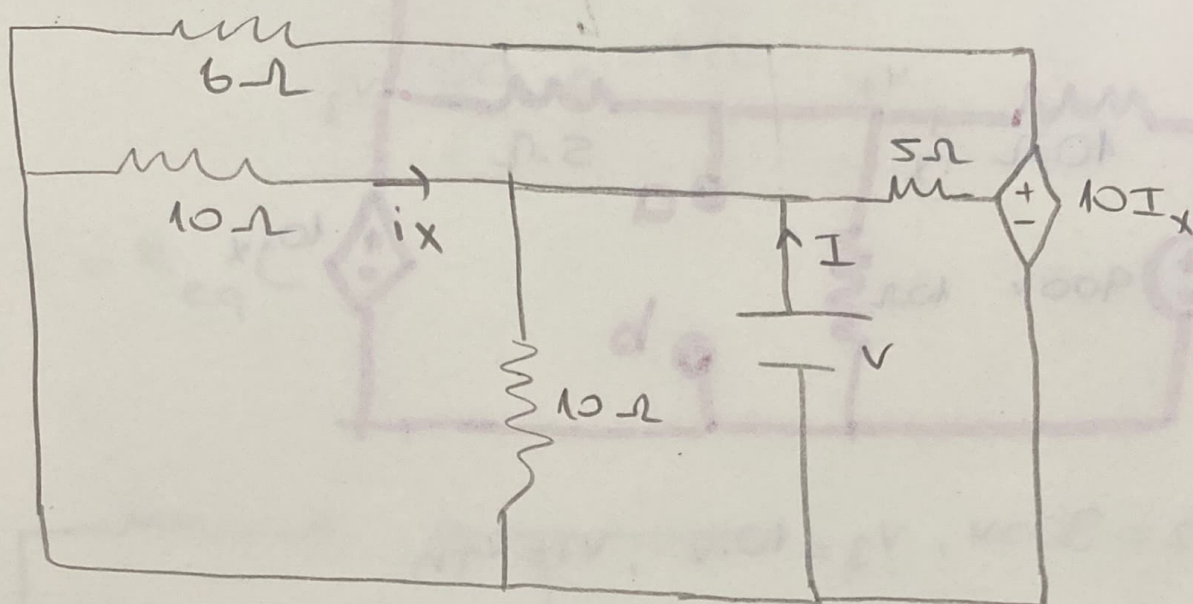
$$4V_1 - 2(900 - V_1) = 900 \rightarrow 4V_1 - 1800 + 2V_1 = 900$$

$$6V_1 = 2700$$

$$V_1 = 450$$

$$V_1 = V_{Th} = 450\text{V}$$

deactive independent sources:



$$i_x = \frac{0 - V}{10} = -\frac{V}{10}$$

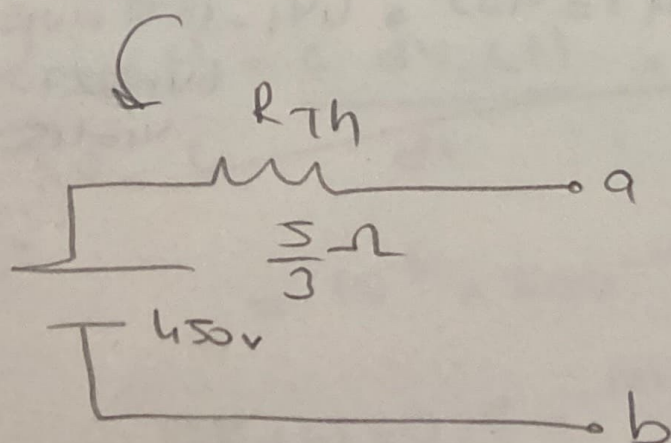
For node - A:

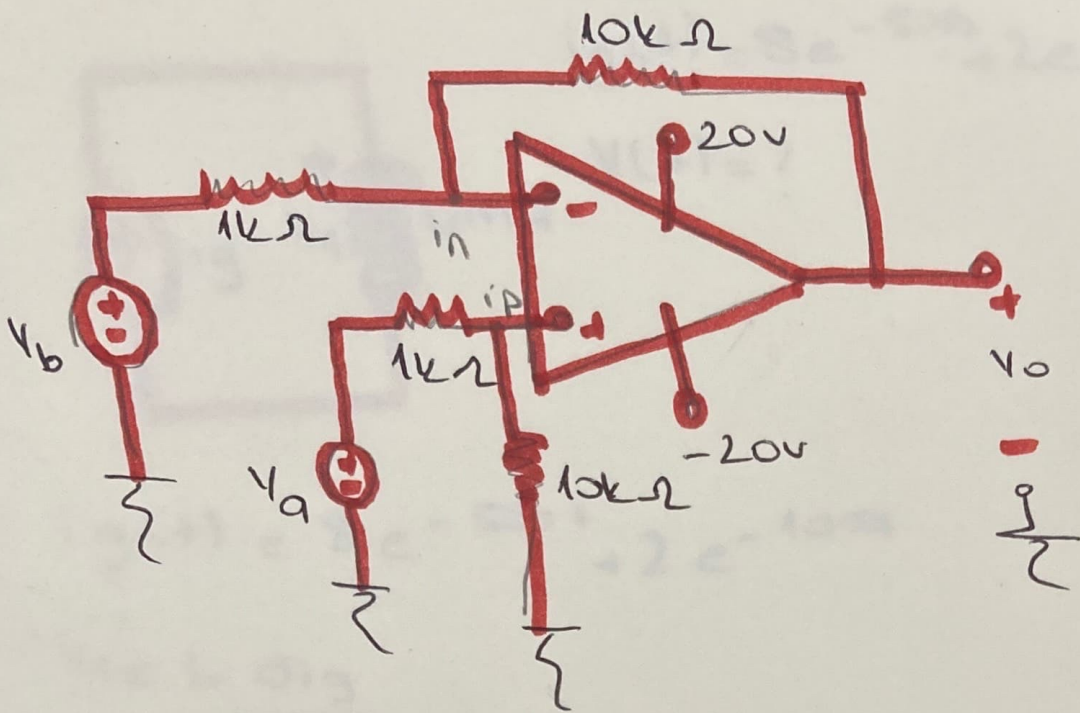
$$\frac{V}{10} + \frac{V}{10} + \frac{V - 10I_x}{5} = I$$

$$V + V + 2V - 20I_x = 10I$$

$$6V = 10I$$

$$\frac{V}{I} = \frac{5}{3} \Omega = R_{Th}$$





$$i_P = i_N = 0, \quad V_N = V_P = V_1$$

$$V_P = \frac{10k\Omega}{10k\Omega + 1k\Omega} V_b = \frac{10}{11} V_b = V_1 \rightarrow \boxed{10V_b = 11V_1} \star$$

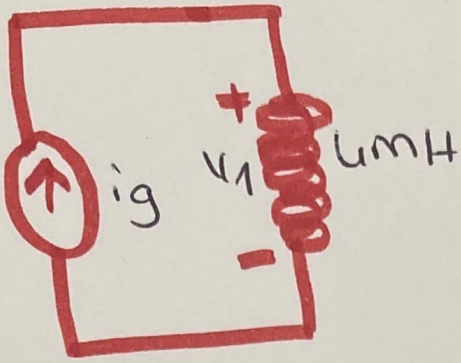
KCL for node -1:

$$\frac{V_1 - V_a}{1k\Omega} + \frac{V_1 - V_o}{10k\Omega} = 0 \rightarrow 10V_1 - 10V_a + V_1 - V_o = 0$$

$$11V_1 - 10V_a = V_o$$

$$11V_1 = 10V_b$$

$$\boxed{10V_b - 10V_a = V_o} \star$$



$$i_g(t) = 8e^{-500t} + 2e^{-1000t}$$

$$v(t) = ?$$

$$i_g(t) = 8e^{-500t} + 2e^{-1000t}$$

$$v_1 = L \frac{di_g}{dt}$$

$$= (4 \times 10^{-3}) \cdot [8(-500)e^{-500t} + 2(-1000)e^{-1000t}]$$

$$\text{Power delivered} = \int v i dt$$

$$= \int L \cdot \frac{di_g}{dt} \cdot i_g \cdot dt$$

$$= \int L \cdot i_g \cdot d i_g$$

$$= L \cdot \frac{i_g^2}{2}$$

$$= 2 \times 10^{-3} [8e^{-500t} + 2e^{-1000t}]^2 //$$